


Figure 9: **Harness optimization is compatible with fine-tuning.** On Qwen-Image-Edit-2511, AutoRef-Harness outperforms DyRef alone, and applying it to the DyRef-trained model yields a further improvement. Ours denotes AutoRef-Harness.

Table 10: OmniContext, SINGLE and MULTIPLE task types, 15 tasks each: prompt following, subject consistency, and their geometric mean behind Table 3. *Gen.* is images drawn per task. Best open image generators per column in bold, second best underlined.

| | | SINGLE | | | | | | MULTIPLE | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Character | | | Object | | | Character | | | Object | | | Char. + Obj. | | |
| Method | Gen. | PF | SC | Overall | PF | SC | Overall | PF | SC | Overall | PF | SC | Overall | PF | SC | Overall |
| GPT-Image-1.5 | 1 | 9.80 | 9.33 | 9.56 | 9.80 | 9.60 | 9.70 | 9.67 | 9.00 | 9.32 | 9.67 | 9.27 | 9.46 | 9.27 | 9.27 | 9.26 |
| Nano Banana Pro | 1 | 9.53 | 9.73 | 9.63 | 9.53 | 9.33 | 9.42 | 9.73 | 9.20 | 9.46 | 9.40 | 9.00 | 9.19 | 9.00 | 9.07 | 9.02 |
| Seedream 4.5 | 1 | 9.60 | 9.33 | 9.45 | 9.47 | 9.60 | 9.50 | 9.20 | 9.13 | 9.09 | 9.67 | 9.13 | 9.39 | 9.13 | 9.07 | 9.09 |
| OmniGen2 | 1 | 8.20 | 8.93 | 8.51 | 6.87 | 6.40 | 5.73 | 7.13 | 5.87 | 6.30 | 7.67 | 5.87 | 6.58 | 7.80 | 7.67 | 7.71 |
| DreamOmni2 | 1 | 7.53 | 8.33 | 7.81 | 7.20 | 6.60 | 6.72 | 5.07 | 5.13 | 4.80 | 7.00 | 7.40 | 7.07 | 6.73 | 5.53 | 5.92 |
| BAGEL | 1 | 8.20 | 6.40 | 6.67 | 6.67 | 8.53 | 7.09 | 4.27 | 3.07 | 3.43 | 7.07 | 6.80 | 6.74 | 7.20 | 7.20 | 7.08 |
| FLUX.2 [klein] 4B | 1 | 9.40 | 9.07 | 9.22 | 8.27 | 8.87 | 8.16 | 8.00 | 7.93 | 7.91 | 8.87 | 7.73 | 8.21 | 8.47 | **8.93** | 8.68 |
| + Best-of-3 | 3 | 9.40 | **9.13** | **9.26** | 8.60 | 8.20 | 7.97 | 8.73 | 8.73 | 8.71 | 8.87 | 8.87 | 8.81 | 8.73 | 8.87 | 8.79 |
| + **AutoRef-Harness** | 3 | 9.33 | 8.60 | 8.94 | 9.20 | **9.00** | 8.99 | 9.33 | 8.73 | 9.02 | 9.00 | 8.60 | 8.78 | 8.53 | 8.67 | 8.59 |
| FLUX.2 [klein] 9B | 1 | 9.40 | 9.00 | 9.17 | 9.33 | 8.93 | 9.08 | 9.00 | 8.40 | 8.66 | 8.73 | 7.73 | 8.14 | **8.87** | 8.67 | 8.74 |
| + **AutoRef-Harness** | 3 | **9.47** | 9.00 | 9.22 | 9.53 | 8.87 | 9.18 | **9.40** | **9.00** | **9.19** | **9.73** | **8.93** | **9.32** | 8.80 | 8.87 | **8.82** |
| Qwen-Image-Edit-2511 | 1 | 9.40 | 8.93 | 9.14 | **9.73** | 8.73 | **9.19** | 8.87 | 8.53 | 8.66 | 9.60 | 8.47 | 9.00 | 8.20 | 8.53 | 8.33 |
| + **AutoRef-Harness** | 3 | 9.27 | 9.00 | 9.12 | 9.13 | 8.47 | 8.64 | 9.33 | 8.87 | 9.09 | 9.13 | 8.20 | 8.62 | 8.47 | 8.27 | 8.34 |

Table 11: OmniContext, SCENE task types. Columns as in Table 10.

| | | SCENE | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Character | | | Object | | | Char. + Obj. | | |
| Method | Gen. | PF | SC | Overall | PF | SC | Overall | PF | SC | Overall |
| GPT-Image-1.5 | 1 | 10.00 | 9.40 | 9.69 | 9.53 | 9.27 | 9.39 | 8.80 | 9.13 | 8.93 |
| Nano Banana Pro | 1 | 9.73 | 9.00 | 9.35 | 8.07 | 8.87 | 8.39 | 7.93 | 8.53 | 8.20 |
| Seedream 4.5 | 1 | 9.87 | 8.87 | 9.35 | 8.60 | 8.80 | 8.66 | 8.13 | 8.47 | 8.23 |
| OmniGen2 | 1 | 7.27 | 6.80 | 6.93 | 6.47 | 5.87 | 6.10 | 7.53 | 6.73 | 7.03 |
| DreamOmni2 | 1 | 6.40 | 5.40 | 5.78 | 6.40 | 5.07 | 5.63 | 6.20 | 5.40 | 5.72 |
| BAGEL | 1 | 4.87 | 4.00 | 3.97 | 4.13 | 4.20 | 4.11 | 5.73 | 5.67 | 5.63 |
| FLUX.2 [klein] 4B | 1 | 9.67 | 8.73 | 9.18 | 7.40 | 7.67 | 7.47 | 7.40 | 7.73 | 7.54 |
| + Best-of-3 | 3 | 9.73 | **8.93** | 9.32 | 8.73 | 8.33 | 8.49 | 7.93 | 7.80 | 7.74 |
| + **AutoRef-Harness** | 3 | 9.73 | 8.87 | 9.28 | 9.00 | **8.80** | **8.89** | 8.27 | **8.33** | 8.28 |
| FLUX.2 [klein] 9B | 1 | **9.87** | 8.87 | **9.35** | 7.87 | 8.47 | 8.07 | 7.27 | 7.80 | 7.47 |
| + **AutoRef-Harness** | 3 | **9.87** | 8.87 | **9.35** | 8.80 | 8.07 | 8.40 | **8.60** | 8.20 | **8.30** |
| Qwen-Image-Edit-2511 | 1 | 7.67 | 6.60 | 6.97 | 8.40 | 8.07 | 8.17 | 8.53 | 7.87 | 8.17 |
| + **AutoRef-Harness** | 3 | 9.07 | 8.40 | 8.71 | **9.27** | 8.40 | 8.82 | 8.40 | 8.20 | 8.28 |